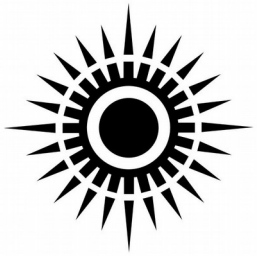
SOLARPUNK
PROCESS BOOK



BOARDGAME PROJECT

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DISCOVERY

1) Choice of Medium

Regarding the reasons for choosing games, there are actually three main reasons. First, games are a medium that acts on consciousness. What games provide is essentially an experience. Experience is what we often refer to as empathy, and this is always the most direct way to achieve understanding.

Secondly, on a philosophical level, games are a very suitable medium for us.

"We are, as a species, homoludens, and all perfect social systems evolve within games. Games not only conform to our nature, the rights to play are also a natural expression of fundamental human rights." - Johan Huizinga

We are naturally inclined to explore the world within the mechanics and rules of games. Our entire civilization is actually formed within games. In industrial society, people should pay more attention to games because games are the main channel through which we remain human and overcome the alienation imposed on us by social systems in the industrial society. This is related to the production chains of industrial society, which enhance efficiency through division of labor and mass production lines. This causes people to be confined to a link in the production chain. Such restrictions break the natural logic of feedback acquisition inherent to humans. We humans pursue visible impacts on the surrounding world, including the impact on our social and natural environment. This strong feedback is now hard to

obtain outside of games. Games are our natural utopia.
Finally, the growth potential of the gaming industry is enormous. In recent years, AI technology has developed rapidly, and the productivity brought by these technologies may soon solve the material deficiencies of human life. The demand for spiritual fulfillment will become increasingly emphasized. Entertainment is a key part of this. Please imagine such a scenario: when people no longer have to worry about material conditions, what will they do besides play games?

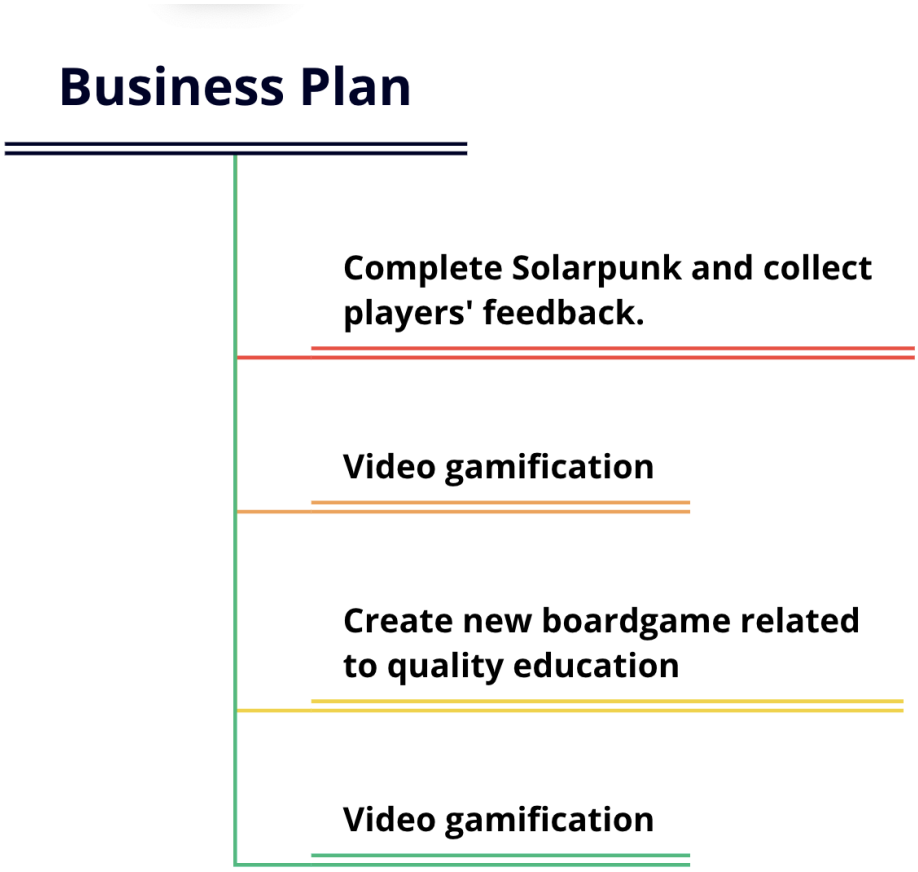
2) Vision

To promote society to get closer to the sustainable development goals through games.

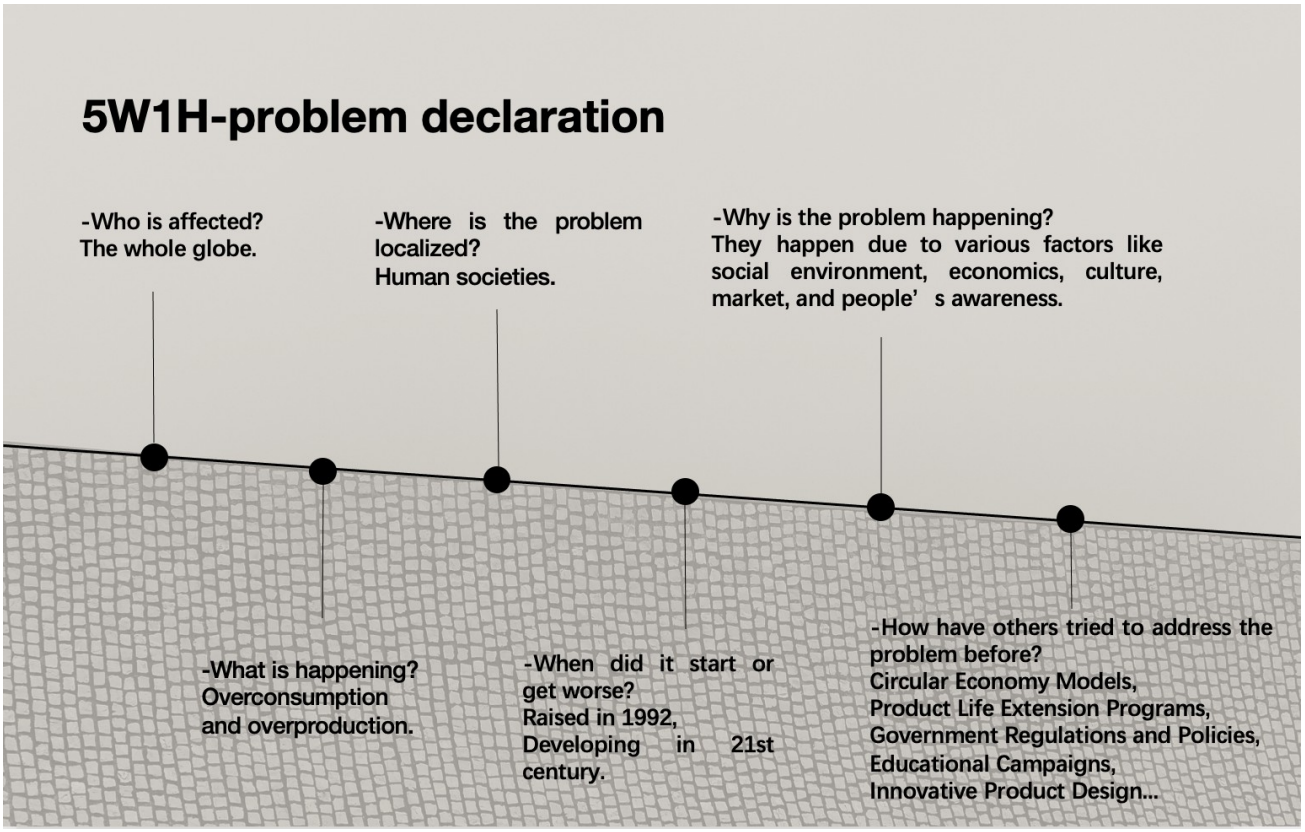
3) Mission

To develop and integrate fun games with sustainable development goals concepts.

4) Business plan



5) The issue and the problem



Problems:

Overconsumption refers to the act of consuming more than what is economically or environmentally sustainable. It occurs when individuals or societies consume resources at a rate that exceeds the planet's ability to regenerate or replace them, leading to various negative consequences such as environmental degradation, resource depletion, and social inequality (Kates et al., 2005).

Overproduction is "the accumulation of unsalable inventories in the hands of businesses, from the tendency of business to expand production beyond the limits of market demand" (O'Connor, 1984).

Reasons behind the problems:

Overconsumption can be attributed to various factors, including **cultural norms and societal pressures, psychological influences, advertising and marketing strategies, and the desire for social status and identity** (Jackson, 2005; Dittmar et al., 2014; Kasser et al., 2016). This innate desire for materialism and status is fulfilled by consuming more than required and it is generally associated with industries related to producing goods that are available in abundance, industries such as the fashion industry producing cheap and affordable clothes in bulk, the more is produced and the more is consumed and it forms an endless loop. These factors contribute to a consumer culture that promotes materialism and encourages individuals to constantly acquire more goods and services.

On the other hand, overproduction is often driven by **economic factors such as the pursuit of profit, market competition, and the need to maintain economic growth** (Harvey, 2014; Foster et al., 2011). Additionally, factors like **technological advancements and globalization** have facilitated increased production capacities and supply chains, contributing to overproduction (Schiederig et al., 2012).

We can include that overconsumption and overproduction happen due to various factors like social environment, economics, culture, market, and people's awareness.

Why do we choose to focus on people's awareness?

1. Structural and Systemic Factors: Overconsumption and overproduction are deeply embedded within the structure and functioning of our economic, social, and political systems. These systems often prioritize economic growth, profit maximization, and short-term gains, which can perpetuate unsustainable consumption and production patterns. Addressing these issues requires systemic changes, such as transitioning to circular economies, evaluating GDP as the primary indicator of progress, and implementing policies that promote sustainable practices throughout supply chains. However, it is hard to realize without the support of the government and the whole market.

2. Industry Practices and Business Models: Many industries operate on models that encourage overconsumption and overproduction. Fast fashion, for example, relies on frequent clothing turnover and low-cost production, which can lead to excessive waste and exploitation of resources and labor. Changing these practices requires transformative shifts in business models, including adopting circular and sustainable approaches, which cannot be done only by developing a project.

3. Policy and Regulation: Government policies and regulations play a crucial role in shaping consumption and production patterns. They can incentivize sustainable practices, discourage waste, and promote responsible production and consumption. Policy interventions may include measures such as taxation, product labelling, extended producer responsibility, and support for sustainable innovation. If we want to address the issue

thoroughly in all aspects, we must ask the government to help us.

Everyone knows that chopping too many trees and using too much water is incorrect. But all serious consequences will only be realized after a long period of time. In other words, excessive production and consumption do not immediately have a negative impact on people's lives, so most people view distant consequences as

having no consequences.

So, the essential problem now is how to change people's AWARENESS of consumption and production.

6) User persona



1. Socially Conscious Student - Max Patel

Background:

-Max Patel is a 21-year-old college student studying environmental science.

-He is passionate about sustainability and actively seeks opportunities to engage with like-minded individuals and projects.

Goals:	Pain Points:
-Max is committed to making a positive impact on the environment and advocating for sustainable practices within his community. -He values activities that allow him to learn, connect, and collaborate with others who share his interests and values. -Max is interested in exploring creative and innovative approaches to addressing environmental challenges and promoting sustainability.	-Max seeks platforms that facilitate meaningful discussions and collaborations around environmental issues, providing opportunities for networking and knowledge-sharing. -He values games that offer a balance of entertainment and intellectual stimulation, challenging players to think critically and creatively. -Max desires opportunities for personal growth and development, engaging in projects and activities that align with his values and goals for the future.

How "Solarpunk" Meets Max's Needs:



-The "Solarpunk" board game's focus on sustainability and responsible consumption resonates with Max's passion for environmental activism and education.

-Max appreciates the game's emphasis on collaboration and problem-solving, fostering meaningful interactions and discussions among players.

-"Solarpunk" provides Max with a platform to connect with fellow students and activists, fostering a sense of community and collective action around important environmental issues.

Background:

-Emily Chen is a
designer with a

-She spends mu
games, board ga
and online gami

Goals:

-Emily seeks gaming experiences that offer rich narratives, engaging gameplay mechanics, and opportunities for creative expression.

-She enjoys participating in gaming events and conventions, where she can connect with fellow enthusiasts and discover new titles.

-Emily values games that provide a sense of escapism and allow her to explore fantastical worlds and characters.

Pain Points:

-Emily is always on the lookout for unique and innovative game concepts that challenge traditional genres and conventions.

-She values games that offer a balance of strategy, social interaction, and thematic immersion, catering to her diverse gaming interests.

-Emily often struggles to find inclusive and welcoming gaming communities where she feels comfortable expressing herself and connecting with like-minded players.

How "Solarpunk" Meets Emily's Needs:

-The "Solarpunk" board game's blend of strategic gameplay and thematic storytelling appeals to Emily's love for immersive gaming experiences.

-Emily appreciates the game's emphasis on social interaction and collaboration, providing opportunities for meaningful player interactions and shared experiences.

-"Solarpunk" offers Emily a fresh and innovative gaming concept that combines her passion for gaming with her interest in sustainability and responsible consumption, making it a standout addition to her gaming collection.

政策项目卡：政策项目会带来很多一次性 “收益”，你不仅可以
研究项目卡：研究项目是一种一劳永逸的卡，花费一次费用但
动的价格交易一份复制过来。

Game Setup: The **global pollution index** starts at 40. The game ends when the **global pollution index** reaches 100, resulting in all players failing; the game also ends if the **global pollution index reaches** 0, at which point the scores for each player's **national development index** are calculated to determine the winner of the game. If a player's **national development index** reaches 100 during the course of the game, the game ends with the winner whose **national development index** reaches to 100.

At the start of the game, four players draw six cards from the **project** deck and keep four as their hand. Each player will also have 50 **money** as start money. Players take turns to take action clockwise. Each turn, every player draw 2 cards from **project** deck. After the player plays the cards, the player need to fold the hand to 4 cards before next player's turn. Players could only play 1 card at most for each kind of **project**. If you want to play another one card in one kind, you need to pay 8 $ for the first one, +8 $ for the laters. Once every player ends their turns, the round ends. There are 4 kinds of cards in **project** deck: **Industrial project** cards, **environmental project** cards, **research project** cards and **policy** cards. **Industrial project** cards earn **money** and increase **national development index** but increase the **global pollution index**, **environmental project** cards decrease the **global pollution index** and **cost money**, **research projects** cards have a continuous effect throughout the game, and **policy** cards affect the other three types of projects. When **industrial project** and **environment poject** are repealed, its effect on **national development index** and **global pollution index** need to be got rid of. Players can also copy and sell **active research projects** to other players, with the **money** negotiated between the players themselves.
At the beginning of the game, draw one card from **mission** deck as a **public mission** and put it into **public mission** area. The first player to complete the **public mission** receives the corresponding **national development index** indicated on the card immediately. At the end of that round, draw 3 cards from **mission** deck and simultaneously **vote** one as the new **public mission**. Players can buy vote in 6 **money** each(put **money** under the **vote** token as purchased **vote**). If there are multiple **missions** getting the same **vote**, the **public mission** are randomly decided.
During the process of the game, if one player's **money** equals to 0 or below, the player needs to exchange **money** with **national development index** in the ratio of 1:1. If a player's **national development index** cannot afford the **money**, the player fail the game immediately.

b) System Analysis

Money System: Money system stimulates the cost of sustainable development, research and policy making. It restricts the frequency of playing cards of each player and bridge other systems up with a constant value.

National Development Index: It is the core mechanism in this game, the winning of this game cannot leave this system because it is the point itself in point-oriented game. It stimulates the concept of development in the real world.

Global Pollution Index: This counter system is to constraint the pace of development and make the whole game more strategic. Also, it induces the thinkings of competition and cooperation of this game among players.

Mission System: Mission System is an auxiliary system for players to gain points. It makes the system more complicated in strategy making, just like Pollution Index, though it is on the other direction. It also enhances the effect of environment projects as extrinsic motivation.

4) Storytelling and Play-flow

a) Storytelling:

The storytelling of our game falls on the process of playing. It contains how players acts during the game, what cards they play, how they communicate with other players. Anyway, the destination of story only has 2 situations: Win and Lose. Still, we have some patterns to assist players' storytelling like visual design, trade agenda, interaction cards designs.

b) Play-flow:



5) Card Design:

library

研究项目 37

绿色发展研讨：-16￥ 抽牌阶段抽到环境牌为止
苏联模式：-16￥抽牌阶段抽到工业牌为止
政策优化：-16￥抽牌阶段抽到政策牌为止
水循环：-10￥每当回合结束时，-2xE （x=本回合弃牌数）
水循环：-10￥每当回合结束时，-2xE （x=本回合弃牌数）
生物能研究：-10￥ 每回合额外抽2张牌
生物能研究：-10￥ 每回合额外抽2张牌
垃圾分类：-10￥ 弃牌阶段可保留6张牌
垃圾分类：-10￥ 弃牌阶段可保留6张牌
碳捕捉：-8￥ 若碳封存或碳中和在激活区，获得所有激活区的环境项目的每回合E（最多15E）
碳封存：-8￥ 若碳中和或碳捕捉在激活区，获得所有激活区的工业项目的N（最多15N）
碳中和：-8￥ 若碳封存或碳捕捉在激活区，获得所有激活区的研究项目的￥（最多30￥）
碳捕捉：-8￥ 若碳封存或碳中和在激活区，获得所有激活区的环境项目的每回合E（最多15E）
碳封存：-8￥ 若碳中和或碳捕捉在激活区，获得所有激活区的工业项目的N（最多15N）
碳中和：-8￥ 若碳封存或碳捕捉在激活区，获得所有激活区的研究项目的￥（最多30￥）
有机农业：-6￥ 每弃一张牌+2￥
有机农业：-6￥ 每弃一张牌+2￥
育种技术研究：-12￥ 每弃一张牌+4￥
育种技术研究：-12￥ 每弃一张牌+4￥
可降解塑料：-10￥ 每个环保项目每回合消耗-2￥/T
可降解塑料：-10￥ 每个环保项目每回合消耗-2￥/T
可持续发展教育：-12￥ 每个政策项目消耗-4￥
可持续发展教育：-12￥ 每个政策项目消耗-4￥
AI项目：-12￥ 每个工业项目每回合收益+4￥/T
AI项目：-12￥ 每个工业项目每回合收益+4￥/T
核研究：-16￥ 后续每个工业项目收益+4N
核研究：-16￥ 后续每个工业项目收益+4N
可持续水产养殖技术：-20￥ 每回合额外抽4张牌
可持续水产养殖技术：-20￥ 每回合额外抽4张牌
绿色供应链优化：-10￥ 你每回合都可以弃任意张牌以抽相同数目的牌（限一次）
绿色供应链优化：-10￥ 你每回合都可以弃任意张牌以抽相同数目的牌（限一次）
卫星技术：-10￥ +3N/T
卫星技术：-10￥ +3N/T
纳米技术：-16￥ +6N/T +2E/T
纳米技术：-16￥ +6N/T +2E/T
国际政治学研究：-15￥ 公共任务得分翻倍
国际政治学研究：-15￥ 公共任务得分翻倍

环境项目 36

人工珊瑚：-7￥/T -8E/T 激活区每有一个研究项目，此项目+2￥/T（最多10￥/T）
人工珊瑚：-7￥/T -8E/T 激活区每有一个研究项目，此项目+2￥/T（最多10￥/T）
国家公园： -6￥/T -10E/T
国家公园： -6￥/T -10E/T
海洋保护区： -5￥/T -8E/T 丢弃所有手牌，抽相同数量的牌
海洋保护区： -5￥/T -8E/T 丢弃所有手牌，抽相同数量的牌
汽车油改电： -5￥/T -4E/T 每个激活的工业项目+5E
汽车油改电： -5￥/T -4E/T 每个激活的工业项目+5E
自然保护区： -6￥/T -6E/T 抽两张牌
自然保护区： -6￥/T -6E/T 抽两张牌
植树造林：-5￥/T -6E/T
植树造林：-5￥/T -6E/T
垃圾分类：-4￥/T -6E/T 抽三张牌，然后选择一张置入你的手牌
垃圾分类：-4￥/T -6E/T 抽三张牌，然后选择一张置入你的手牌
风能补贴：-6￥/T -8E/T 抽2张牌
风能补贴：-6￥/T -8E/T 抽2张牌
水能补贴：-8￥/T -10E/T 本回合结束无需弃牌
水能补贴：-8￥/T -10E/T 本回合结束无需弃牌
太阳能补贴：-6￥/T -8E/T 回合结束时收入大于20￥可额外-4E
太阳能补贴：-6￥/T -8E/T 回合结束时收入大于20￥可额外-4E
可持续建筑设计：-10￥/T -12E/T 抽4张牌
可持续建筑设计：-10￥/T -12E/T 抽4张牌
生态旅游：-6￥/T -5E/T 抽2张牌
生态旅游：-6￥/T -5E/T 抽2张牌
污染土壤修复：-12￥/T -10E/T
污染土壤修复：-12￥/T -10E/T
河流生态恢复：-6￥/T -20E
河流生态恢复：-6￥/T -20E
绿色化学品升级：-4￥/T -16E 若E<=30 获得5N
绿色化学品升级：-4￥/T -16E 若E<=30 获得5N
清洁工业：-5￥/T -16E 若你的N>=50 额外获得10￥
清洁工业：-5￥/T -16E 若你的N>=50 额外获得10￥
水坝：-6￥/T -3E/T 每个激活的工业项目+2N
水坝：-6￥/T -3E/T 每个激活的工业项目+2N
海绵城市：-3￥/T -4E/T 抽两张牌
海绵城市：-3￥/T -4E/T 抽两张牌

任务牌 25

成交2个研究项目：10N
成交1个研究项目：5N
拥有6个环境项目：20N
拥有4个环境项目：15N
拥有3个环境项目：10N
每种牌各使用2张：10N
累计弃6张牌：5N
累计弃12张牌：10N
连续3回合打出政策牌：10N
连续2回合打出政策牌：5N
连续3回合打出环境牌：10N
连续2回合打出环境牌：5N
连续5回合打出环境牌：20N
累计抽8张牌：5N
累计抽13张牌：10N
连续2回合打出研究牌：5N
连续3回合打出研究牌：10N
收入50￥：5N
收入80￥：10N
累计撤除3个工业项目：10N
累计撤除5个工业项目：15N
环境项目花费总额30￥：10N
环境项目花费总额50￥：15N
激活的环境项目中有1组相同的牌：5N
激活的环境项目中有2组相同的牌：15N

工业项目 42

填海造陆：+1￥/T +6E/T +20N
工业区规划：+2￥/T +2E/T +15N，每有一个工业项目+5E
砍树：+10￥/T +3E/T +3N，可最多丢弃3张牌，每张牌可换取4￥，但每张牌+2E
砍树：+10￥/T +3E/T +3N，可最多丢弃3张牌，每张牌可换取4￥，但每张牌+2E
化工产业：+10￥/T +8E/T +10N，每有一个工业项目额外+2N
垃圾填埋：+3￥/T +8E +4N，丢弃x张牌，每丢弃一张牌+1N（x<=6）
垃圾焚烧：+2￥/T +8E +5N，丢弃x张牌，每丢弃一张牌+3￥（x<=6）
垃圾填埋：+3￥/T +8E +4N，丢弃x张牌，每丢弃一张牌+1N（x<=6）
垃圾焚烧：+2￥/T +8E +5N，丢弃x张牌，每丢弃一张牌+3￥（x<=6）
城市化：+2￥/T +9E +3N，购买购买的票不需要￥
矿石开采：+4￥/T +15E +1N
煤炭开采：+2￥/T +6E +5N 抽2张牌
煤炭开采：+2￥/T +6E +5N 抽2张牌
智能制造工厂：+2￥/T +16E +8N 抽3张牌
智能制造工厂：+2￥/T +16E +8N 抽3张牌
重金属冶炼：+6￥/T +12E/T +16N 本回合结束时无需弃牌
重金属冶炼：+6￥/T +12E/T +16N 本回合结束时无需弃牌
化肥厂：+2￥/T +10E +5N
造纸厂：+3￥/T +15E +8N
石油厂：+12￥/T +12E/T +8N 若当前E>=50， 变为+10￥/T
石油厂：+12￥/T +12E/T +8N 若当前E>=50， 变为+10￥/T
皮革厂：+2￥/T +8E +5N
水泥厂：+2￥/T +10E +8N 抽2张牌
水泥厂：+2￥/T +10E +8N 抽2张牌
水坝：+4￥/T +8E/T +10N 每有一个环境项目-2E
水坝：+4￥/T +8E/T +10N 每有一个环境项目-2E
油漆厂：+3￥/T +10E +3N
电镀厂：+4￥/T +8E/T +15N
发电厂：+3￥/T +6E +8N N<=20时才能激活
发电厂：+3￥/T +6E +8N N<=20时才能激活
纺织厂：+3￥/T +8E +4N 抽一张牌
纺织厂：+3￥/T +8E +4N 抽一张牌
养殖厂：+4￥/T +5E/T +10N 弃2张牌
养殖厂：+4￥/T +5E/T +10N 弃2张牌
制药厂：+8￥/T +8E/T +12N N>=60时才能激活
制药厂：+8￥/T +8E/T +12N N>=60时才能激活
军工厂：+8￥/T +10E/T +15N 后续任务不能投票
畜牧业：+3￥/T +8E +4N
电子厂：+4￥/T +4E/T +10N 弃两张牌 然后抽两张牌
重工业区：+12￥/T +15E/T +20N 每回合收益-1￥/T
天然气开发：+6￥/T +1E/T +10N 每回合收益-1￥/T
海绵城市：+3￥/T +2E/T +5N 抽两张牌

政策 44

商业间谍：8￥ 从每个玩家手牌处抽一张牌
商业间谍：8￥ 从每个玩家手牌处抽一张牌
商业间谍+：12￥ 从每个玩家手牌处选一张牌
商业间谍+：12￥ 从每个玩家手牌处选一张牌
禁止开发：6￥ 撤除所有工业项目，但不会减少已经获得的N，每撤除一个工业项目+6￥
禁止开发：6￥ 撤除所有工业项目，但不会减少已经获得的N，每撤除一个工业项目+6￥
核废水入海：10￥ +10E，之后不能购买票，其他玩家每人-5N
核废水入海：10￥ +10E，之后不能购买票，其他玩家每人-5N
休养生息：3￥ 撤除任意环境或工业项目，直到其数量相等
休养生息：3￥ 撤除任意环境或工业项目，直到其数量相等
国际法庭+：15￥ 所有一名已激活的环境项目少于你的玩家，获得其12￥
国际法庭+：15￥ 所有一名已激活的环境项目少于你的玩家，获得其12￥
国际法庭：8￥ 所有已激活的环境项目少于你的玩家，获得其6￥
国际法庭：8￥ 所有一名已激活的环境项目少于你的玩家，获得其6￥
原料进口：14￥ 每有一个环境项目，-4E，+2N
原料进口：14￥ 每有一个环境项目，-4E，+2N
“谁污染谁治理”：5￥ 撤除最多两个环境项目
“谁污染谁治理”：5￥ 撤除最多两个环境项目
工业间谍：10￥ 撤除一个其他人的工业项目
工业间谍：10￥ 撤除一个其他人的工业项目
工业间谍：18￥ 撤除2个其他人的工业项目
工业间谍：18￥ 撤除2个其他人的工业项目
科技间谍：8￥ 复制一个其他人的研究项目
科技间谍：8￥ 复制一个其他人的研究项目
排放审查：6￥ 撤除最多3个工业项目，每撤除一个工业项目+4￥
排放审查：6￥ 撤除最多3个工业项目，每撤除一个工业项目+4￥
限氟令：4￥ 最多撤除两个工业项目
限氟令：4￥ 最多撤除两个工业项目
五年计划：10￥ 撤出最多3个环境项目，你激活的每个工业项目+2N
五年计划：10￥ 撤出最多3个环境项目，你激活的每个工业项目+2N
大跃进：12￥ 撤出所有环境项目，每撤出一个+4N +2E
大跃进：12￥ 撤出所有环境项目，每撤出一个+4N +2E
科教兴国：8￥ 本回合下一个研究项目免费
科教兴国：8￥ 本回合下一个研究项目免费
世界议题：5￥ 你的回合结束后跳过该任务进行重新投票
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世界议题：5￥ 你的回合结束后跳过该任务进行重新投票
气候会议+：16￥从工业项目最多的玩家获得20￥，最多环境项目玩家+8N
气候会议+：16￥从工业项目最多的玩家获得20￥，最多环境项目玩家+8N
气候会议：10￥从工业项目最多的玩家获得12￥，最多环境项目玩家+4N
气候会议：10￥从工业项目最多的玩家获得12￥，最多环境项目玩家+4N
科研会议：12￥复制场上的任意一个研究送给所有人
科研会议：12￥复制场上的任意一个研究送给所有人

6) Visual Design:



a) Research Card:

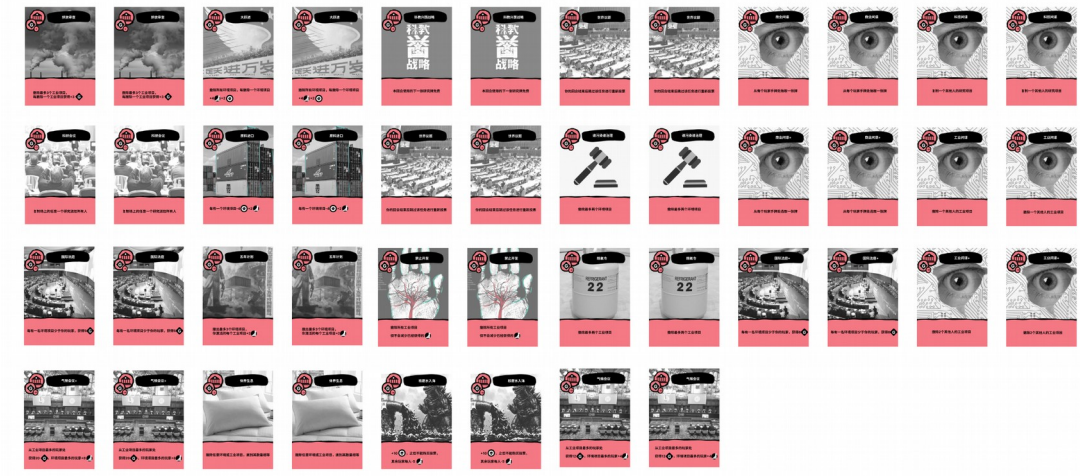
b) Environment Project Card:



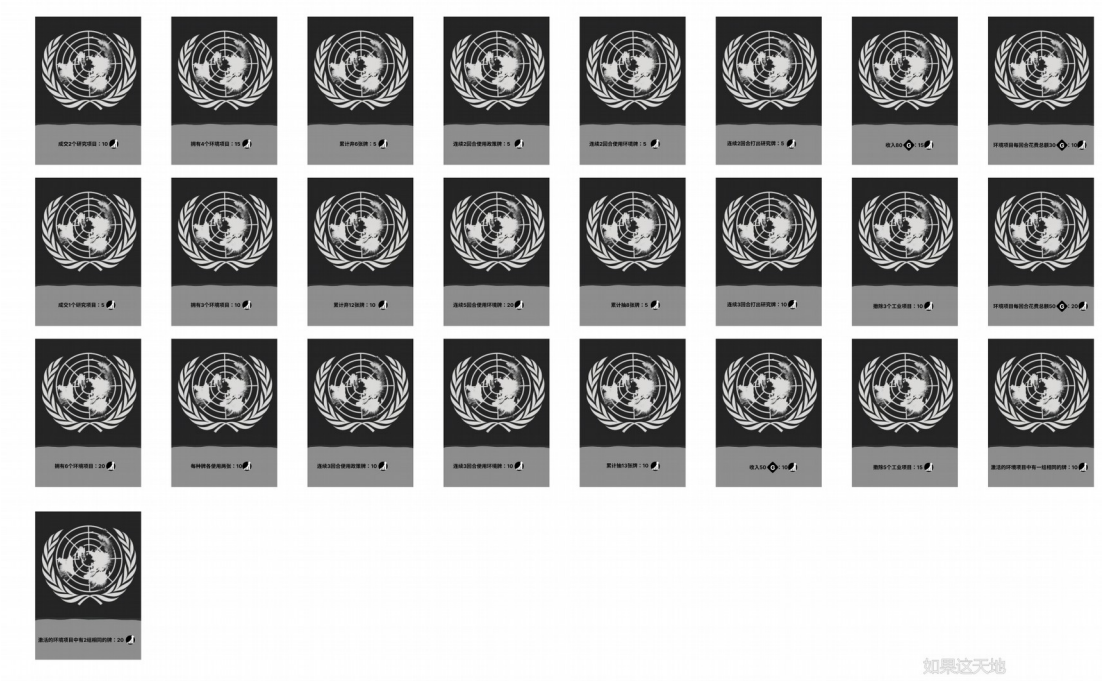
c) Industry Project Card:



d) Policy Card



e) Mission Card



f) Board


已激活的项目
SOLAR
PUNK
85
80
75
70
65
60
55
50
45
40
35
30
25
20
15
10
5
0
90
95
100



(All Project File is in This Link: https://www.figma.com/file/ETqiH90WUAhbk3aN5k6kOW/Untitled?type=design&node-id=0%3A1&mode=design&t=ldMQXSCj9I1ho2tb-1)

7) Playtest

a) Alpha Test:

α1.1 log: Designer Internal Test1

We failed to finish the game in the first test.

Basic mechanism of game has some problems.

We adjust:

1. The limitation of playing cards each turn (from once each kind to pay more money to play more cards).
2. Start money (30 to 50)

3. Ratio of different kinds of project (2:2:3:2 to 1:1:1:1)
4. Mechanism: some of Affect per turn to Affect once
5. Some weak cards change on Value

α1.2 log: Designer Internal Test2
This time we finished the game with 4 failures on >100 global pollution index and 1 win on National Development Index. It is really hard to win the game and hard to operate the number, so we made some adjustment:
1. Global pollution index starts from 30
2. Use each player's turn settlement rather than 4 players' turn settlement.
3. Some cards are weak.
4. More mission cards on environment project
5. Revote of mission
6. Add more interaction cards like steal, remove others' project, copy project…
7. Win settlement.

b) Beta Test

β1.1 log: Public test
We invite some professional players of boardgame to this test. Their feedback shows that the overall experience is good but some of cards are too IMBA and some are useless. We made some adjustment below:
1. Global pollution index starts from 40
2. Adjustment on the value of some cards
3. Design more Industrial cards and policy cards
4. Dice decision on equal vote
5. Buying vote from 5$ to 6$
6. Price of playing more cards from 5$ to 8$

7. Board Redesign

DEVELOPMENT AND DELIVERY

1) Business Model

1. Key Partners:

-Game Designer: Partnering with experienced game designers who can create engaging and educational board games aligned with sustainable knowledge themes.
-Education Agency: Collaborating with educational institutions to integrate board games into their curriculum or extracurricular activities, offering workshops and training sessions.
-Social Media Content Creators (COLs - Content Opinion Leaders): Partnering with influencers or content creators in the gaming and education niche to promote the board games and workshops to their followers.
-Manufacturing Company: Partnering with a manufacturing company specializing in board game production to ensure high-quality and cost-effective manufacturing processes.

2. Key Activities:

-Product Design: Constantly innovate and design new board games that are both entertaining and educational, incorporating feedback from users and experts.
-Marketing: Develop comprehensive marketing strategies to promote the board games and workshops, utilizing various channels and techniques.

3. Value Propositions:

-Produce Interesting Board Games: Creating board games that are not only entertaining but also offer valuable learning experiences related to sustainability.
-Gamification for Learning: Utilizing gamification principles to make learning about sustainability engaging and effective.
-Socialization Opportunities: Providing opportunities for social interaction and bonding through gameplay, fostering a sense of community among players.

4. Customer Relationships:

-Education Agencies: Building relationships with education agencies as sellers, buyers, and collaborators, offering them tailored solutions and support in implementing educational board games.
-Game Lovers: Engaging with board game enthusiasts as sellers, buyers, and influencers, encouraging them to spread the word about the games and participate in community events.

5. Customer Segments:

-B2B:

Education Agencies: Targeting educational institutions and agencies looking to enhance their teaching methods with interactive and engaging tools.
Entertainment Spaces (e.g., Playrooms): Partnering with venues that offer recreational activities to provide them with unique and educational board game options.

-B2C:

Board Game Lovers: Catering to individuals who enjoy playing board games, especially those interested in sustainability and educational gaming experiences.
Non-profit environmental organization：Provide NGOs with the product to enrich their influential methods.

6. Key Resources:

-Game Design Skills: Employing skilled game designers capable of creating innovative and educational board games.
-Marketing Techniques: Utilizing various marketing strategies and tools to promote the board games effectively.

7. Channels:

-Email Campaigns: Sending targeted email campaigns to potential partners, customers, and collaborators.
-Workshops: Hosting workshops and training sessions at educational institutions, entertainment venues, and community events.
-Expos: Participating in gaming and education expos to showcase the board games and interact with potential customers.
-Social Media Advertising and Live Streaming: Leveraging social media platforms for advertising, live streaming gameplay sessions, and engaging with the community.
-Own Community (Private Domain): Creating and nurturing an online community platform for board game enthusiasts to interact, share experiences, and provide feedback.

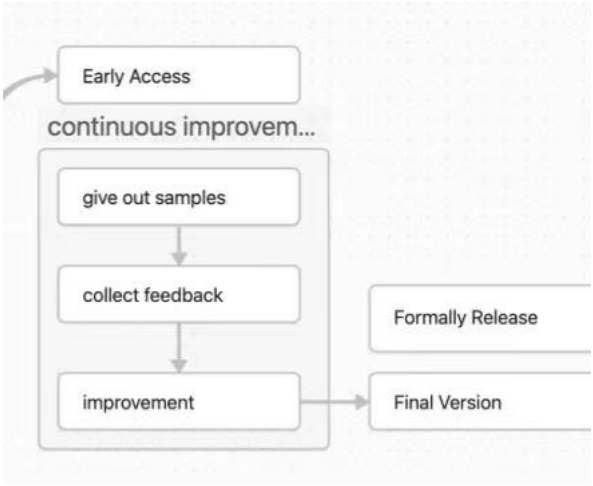
8. Cost:

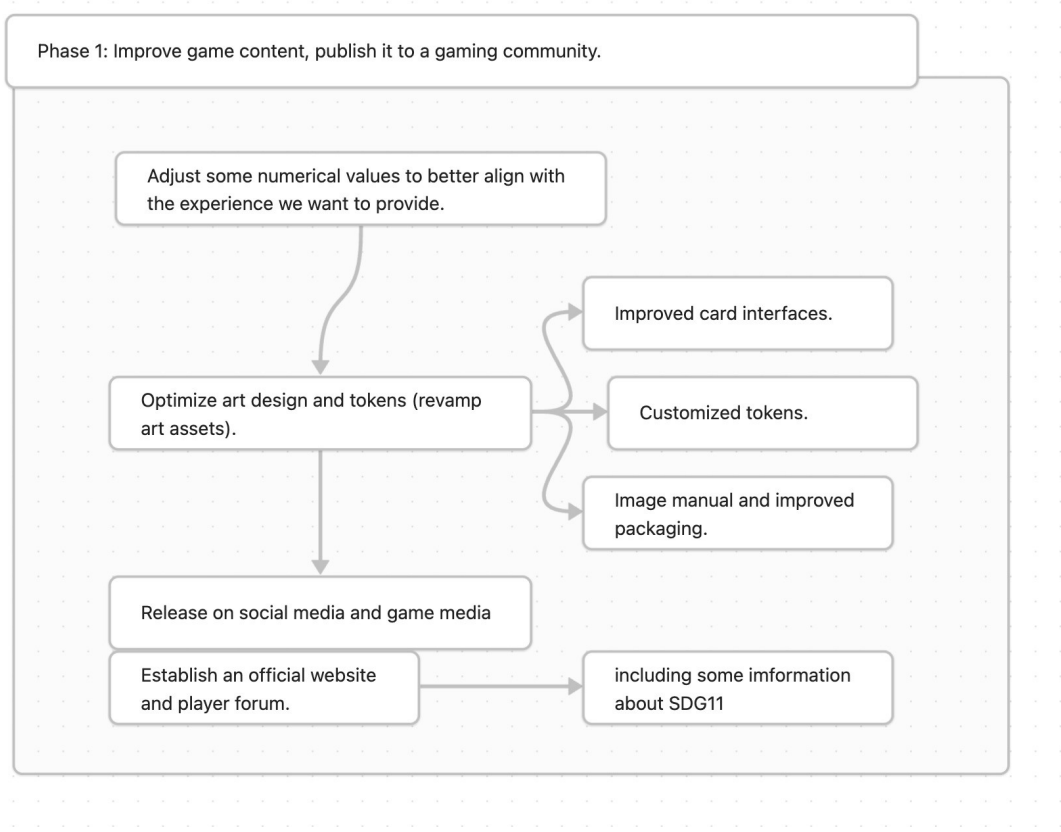
-Production Expenses: Covering the costs associated with designing, prototyping, and manufacturing the board games.
-Marketing: Allocating funds for marketing initiatives such as expos, workshops, social media advertising, and influencer collaborations.
-Employee Salaries: Compensating employees involved in game design, marketing, sales, and customer support.
-Maintenance Fees: Budgeting for ongoing maintenance and updates to the board games and associated platforms.

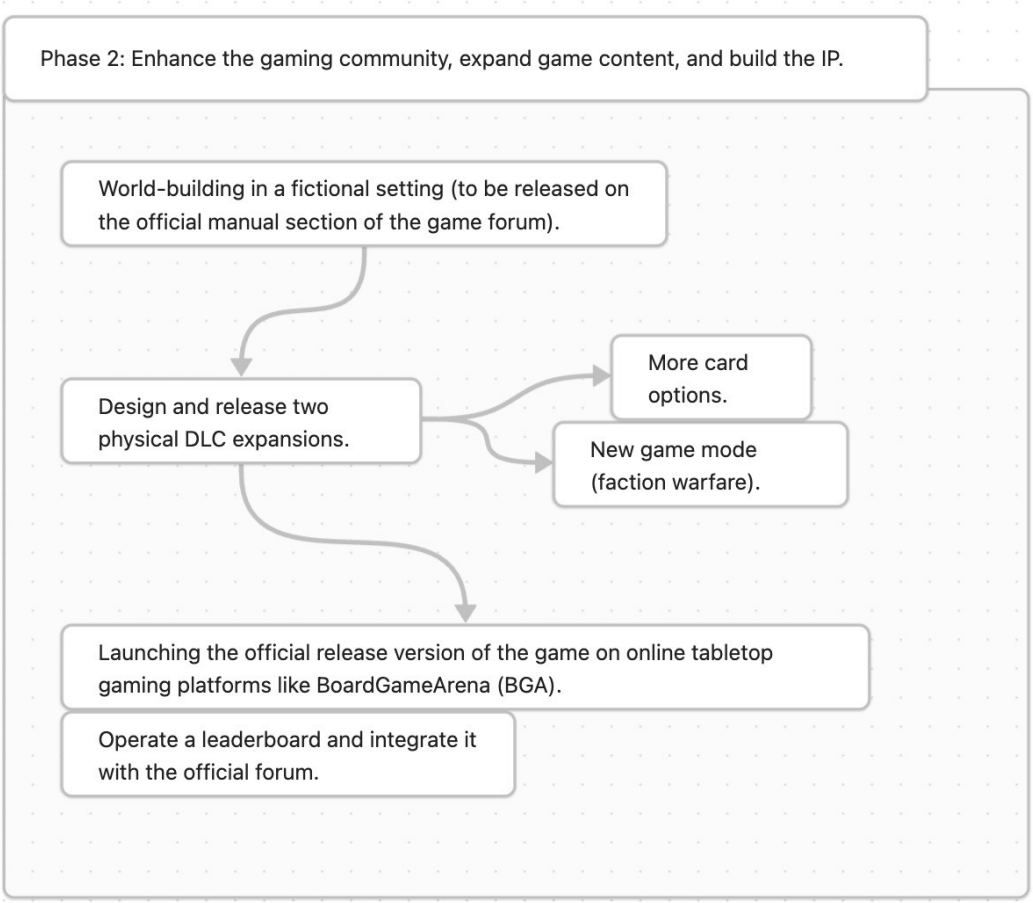
9. Revenue Streams:

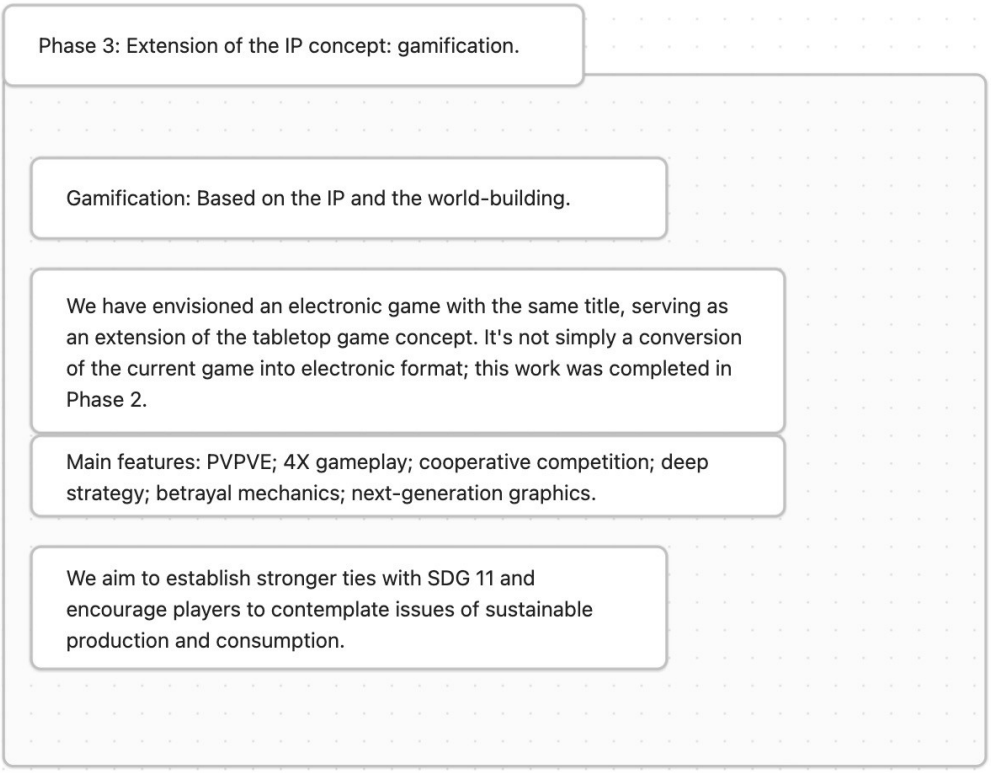
-Game Sales: Generating revenue through the sale of board games to individual consumers, educational institutions, and entertainment venues.
-Workshop Fees: Charging fees for participation in educational workshops and training sessions.
-Licensing Agreements: Entering into licensing agreements with other entities interested in using or distributing the board games.

2) Milestone


Early Access
continuous improvem...
give out samples
collect feedback
improvement
Formally Release
Final Version


Phase 1: Improve game content, publish it to a gaming community.
Adjust some numerical values to better align with the experience we want to provide.
Optimize art design and tokens (revamp art assets).
Improved card interfaces.
Customized tokens.
Image manual and improved packaging.
Release on social media and game media
Establish an official website and player forum.
including some imformation about SDG11


Phase 2: Enhance the gaming community, expand game content, and build the IP.
World-building in a fictional setting (to be released on the official manual section of the game forum).
Design and release two physical DLC expansions.
More card options.
New game mode (faction warfare).
Launching the official release version of the game on online tabletop gaming platforms like BoardGameArena (BGA).
Operate a leaderboard and integrate it with the official forum.


Phase 3: Extension of the IP concept: gamification.
Gamification: Based on the IP and the world-building.
We have envisioned an electronic game with the same title, serving as an extension of the tabletop game concept. It's not simply a conversion of the current game into electronic format; this work was completed in Phase 2.
Main features: PVPVE; 4X gameplay; cooperative competition; deep strategy; betrayal mechanics; next-generation graphics.
We aim to establish stronger ties with SDG 11 and encourage players to contemplate issues of sustainable production and consumption.

References :

Kates, R. W., Parris, T. M., & Leiserowitz, A. (2005). What is sustainable consumption? Journal of Consumer Policy, 28(1), 5-24. doi:10.1007/s00630-005-0012-z

O'Connor, J. (1984). Accumulation Crisis. New York: Basil Blackwell